Week 2

Introduction to LLMs and Reflective Writing

Learning Outcomes:

By the end of this session, students should:

- Know how to write an effective reflective report.
- Be able to apply problem solution structures in reports and reflective writing.
- Be able to report results using appropriate tenses and passive voice.
- Understand some concepts about LLMs.
- Have practiced the pronunciation of some consonant clusters.

Task 1 - Lead-In

In your groups discuss the following questions.

1. What types of writing assignments and assessments did you have to write in year one?
2. What types of writing assignments and assessments will you have to write this year?
3. What do you understand by the term 'reflective writing'?
4. Have you ever done any reflective writing before? How did it go? What did you do well and what did you do badly?

The 'What? - So What? - Now What?' Model.

When reflecting on a project, one of the simplest frameworks you can use is a set of three questions.

What? – What did you do and what happened when you did it?
So what? – What was the importance of what happened? If things went wrong, why?
Now what? – How would you do things different the next time you have to do a similar project?

Task 2 – Reading

Below are three sections of a model report of the reflective report you will be asked to write in week 9. It is divided into 'what?', 'So what?' and 'Now what?' sections. We will look at each one separately.

Task 2a – What?

Read the text below and answer the questions.

(Paragraph 1) During Practical Task 3 of our AI app design project, my responsibility was to evaluate and recommend a suitable AI platform for our mental health chatbot aimed at university students. The task took place over a two-hour lab session, during which I conducted a comparative review of GPT-4, Google Dialogflow, and IBM Watson, based on four criteria: natural language capabilities, ease of integration, ethical considerations, and data privacy features.

(Paragraph 2) I created a comparison table and presented initial findings to my team. We discussed the pros and cons of each platform. While GPT-4 offered the most advanced language generation, we were concerned about the lack of transparent control over outputs. Dialogflow offered a more structured, rule-based design that seemed safer for sensitive conversations. Ultimately, we decided to recommend Dialogflow as the foundation for our chatbot's interaction model.

1. What is the purpose of paragraph 1?
2. What is the purpose of paragraph 2?
3. Does the writer mention anything which went wrong?
4. Does the writer mention how they felt about their project or task?

Task 2b - So What?

Read the text below and answer the questions.

(Paragraph 3) This task made me aware of how important early-stage platform selection is in shaping both functionality and ethical boundaries. Initially, I assumed the most advanced AI (GPT-4) would be the best choice, but our discussion revealed that in contexts like mental health, predictability and safety may matter more than generative power.

(Paragraph 4) I also learned that platform features can constrain or support ethical decision-making. For example, Dialogflow allows developers to define fallback responses and clearly direct users to human support when needed. This helped us conceptualise a safer user journey, reinforcing the need to consider not only what a platform can do, but what it should be allowed to do in context.

(Paragraph 5) This was the first time I applied ethical design principles—like user safety and transparency—to a technical decision. The task helped bridge theory and practice in a very concrete way, even without writing any code.

5. What is the purpose of paragraph 3?
6. What is the purpose of paragraph 4?
7. What is the purpose of paragraph 5?

Task 2c - Now What?

Read the text below and answer the questions.

(Paragraph 6) In future projects, I'll approach platform selection more critically, thinking not only about performance but also about user needs and ethical implications. I plan to improve my ability to assess third-party tools more efficiently and develop a checklist that includes both technical and social criteria.

(Paragraph 7) Next time, I would also push for quick user validation—perhaps a short feedback session with peers—to check whether our assumptions about safety and clarity align with how potential users feel.

(Paragraph 8) This short but focused task showed me how even one design decision can shape an entire app's direction. It made me more aware of the broader responsibilities involved in AI system design.

8. What is the purpose of paragraph 6?
9. What is the purpose of paragraph 7?
10. What is the purpose of paragraph 8?

Task 2d – Wrapping-Up

Now look again at the whole report and answer the following questions.

11. Did the student make any major mistakes during the project?
12. Did the group have any major arguments during the project?
13. Does the writer express any strong emotions they felt during the project?

Task 2e – Verbs in Reflective Reports

Read through each of the three sections again and quickly write down any verbs that describe what the individual student or group of students did.

Verbs

Task 2f – Vocabulary Checking

You probably understood most of the verbs you filled in the table in the last task (2e) with. Some you may be less familiar with. With a friend, look up any you are not sure about in the dictionary then use the most appropriate one to fill in the sentences below.

1. That company has just _____revealed_____its new AI-powered chatbot to the public.

2. We _____conducted_____ research to find out what the public's general opinions about artificial were.
intelligence
3. There have been many news stories recently about Generative AI giving harmful, offensive or highly misleading information to children recently. This only ___reinforces_____ the need for appropriate guardrails.

4. The wide-spread use of generative AI will undoubtedly change our lives in many ways. It is hard, however, for most people to fully __conceptualise_____how the job market will be completely revolutionized
by its introduction.

5. A major discussion in education at the moment ____invovles__concerns_____how to help AI ethically and effectively without simply allowing them to 'cheat' on assessments.
students use generative

The Passive Voice

You will have studied the passive voice before at high school. This section will talk about how it might be used in STEM writing and will remind you of how to produce a passive voice phrase or sentence. Look at the following very simple example:

The cat ate the fish. - (Active voice)

The fish was eaten by the cat. - (Passive voice)

Many students of English understand the passive voice when they read it in a sentence but do not produce it often in their own writing. This is probably because any passive sentence can also be written as an active sentence which is easier to write correctly. Note that both sentences above have exactly the same meaning. The passive voice only reorganizes the words in the sentence so the original object is at the front of the verb rather than the back.

How to make the Passive Voice

The basic form of the passive voice is:

to be + past participle

Remember that the past participle of any verb is the last of the common list of forms that it has - eat, eats, ate, eating and **eaten.** The steps to making the passive voice are as follows.

- Take any tense or subject verb/agreement information from the original verb and move it onto the verb 'to be'. So, for example if your original sentence is past tense, the verb 'to be' should now be past tense. Some examples are:

  ate -> was (past tense -> past tense)
  will eat -> will be (future tense -> future tense)
  could eat -> could be (modal verb -> modal verb)
  is eating -> is being (continuous tense -> continuous tense)
  have eaten -> have been (perfect tense -> perfect tense)

- Put the original verb back into the sentence as the past participle.

  ate -> was **eaten**
  will eat -> will be **eaten**
  could eat – could be **eaten**
  is eating -> is being **eaten**
  have eaten -> have been **eaten**

- Check to see if you need to correct the subject-verb agreement of the verb phrase according to the new subject of the sentence. For example:

  The monster eats the child -> The child <u>is</u> eaten by the monster.

  The monster eats the children -> The children <u>are</u> eaten by the monster.

- Notice that you can sometimes have multiple verbs in the same passive voice phrase.

The cat <u>caught and ate</u> the mouse.

The mouse <u>was caught and eaten</u> by the cat.

Task 3a

Turn the following sentences from active to passive including the original subject using the word 'by'. You can look up the past participle of the verb on the Internet or in your dictionaries if you need to.

1. The businessman sold the house.

2. Shakespeare wrote Hamlet

3. The elephant drank the water.

4. The teacher often blamed the boy.

5. She is taking the exam next week.

6. The President has agreed the plan.

7. By next year, the company will have been selling the product for over one-hundred years

When to Use the Passive Voice.

The passive voice can be used in a number of situations. In STEM writing, it is often used because what happened is often more important that who did it when conducting experiments.

Sometime between 1589 and 1592, Galileo is said to have dropped a feather and a hammer from top of the Leaning Tower of Pisa.

A feather and a hammer were dropped from a height of 60 meters.

The first sentence is more commonly in a **history** book about Galileo, explaining who did what and when. In **scientific** writing, the point is often that the same thing will happen no matter who does it or when. Imagine you are writing the following sentence in your report this semester.

I programmed the app to give three different answers to any question.

The app was programmed to give three different answers to any question.

We already know from the rest of the report that you are the programmer, so we already know you are the one who did it. Therefore, in your **main report**, the second sentence is probably better as it focuses on the app rather than you, the student. However, if you are writing your **reflective report**, you might write something like:

I found that the answer given by the app did not always contain enough information. Therefore, after thinking about it, I programmed the app to give three different answers to any question.

In a reflective report the focus tends to be more on <u>what you did</u> and therefore using the active voice might be better here.

Note that traditionally, students were sometimes told not to use 'I' or 'we' in a scientific or engineering paper. In fact, in real papers, academics do use 'I' or 'we'. However, it is usually better to avoid using these words if there is not a good reason to keep them.

Task 3b

Words	**Explanations**
1. Epoch	a. a stage of data preparation where raw data is cleaned, normalized, and transformed into a format suitable for a machine learning model.
2. Labelled data	b. a neural network that understands the input sequence and compresses its information into a fixed-length contextualized representation.
3. Unlabelled Data	c. data consists of raw input examples without any corresponding annotations or labels.
4. Embedding	d. a transfer learning technique where a pre-trained model is further trained on a smaller, task-specific dataset.
5. Fine-tuned	e. the process of breaking down a sentence or paragraph into smaller pieces, like words or parts of words.
6. Preprocessing	f. a class of transformer models characterized by their massive scale, encompassing hundreds of billions of parameters.
7. Decoder	g. a neural network that generates the output sequence token-by-token, using the contextual representation from the encoder.
8. Encoder	h. Numerical representations of words or concepts where similar items have similar numbers.
9. Tokenization	i. a single, complete pass of a training algorithm over the entire training dataset.
10. Large Language models	j. data consists of input examples that are paired with corresponding ground-truth outputs or annotations.

Task 3c

Look at the extract below which has been adapted from 'Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond' by Yang et. al., an extra reading from week 1. The passage described some of their observations about the evolution of Large Language Models (LLM). Read the passage and highlight any uses of the passive voice. Do not worry if you do not understand every sentence but focus on identifying the grammatical structure. The first one has been done for you.

a) Decoder-only models have been gradually dominating the development of LLMs. At the early stage of LLMs development, decoder-only models were not as popular as encoder-only and encoder-decoder models. However, after 2021, with the introduction of game-changing LLMs - GPT-3, decoder-only models experienced a significant boom. Meanwhile, after the initial explosive growth brought about by BERT, encoder-only models gradually began to fade away

b) Meta contributes significantly to open-source LLMs and promotes research of LLMs. When considering contributions to the open-source community, particularly those related to LLMs, Meta stands out as one of the most generous commercial companies, as all the LLMs developed by Meta were open-sourced.

c) OpenAI consistently maintains its leadership position in LLM, both currently and potentially in the future. Other companies and institutions are struggling to catch up with OpenAI in developing models comparable to GPT-3 and the current GPT-4. This leadership position may be attributed to OpenAI's steadfast commitment to its technical path, even when it was not widely acknowledged initially.

d) LLMs exhibit a tendency towards closed-sourcing. In the early stages of LLM development (before 2020), the majority of models were open-sourced. However, with the introduction of GPT-3, companies have increasingly opted to close-source their models, such as PaLM, LaMDA, and GPT. Consequently, it has become more difficult for academic researchers to conduct experiments on LLM training. As a result, API-based research could become the predominant method in the academic community. No passive voice present.

e) Encoder-decoder models remain promising, as this type of architecture is still being actively explored, and most of them are open-sourced. Google has made substantial contributions to open-source encoder-decoder architectures. However, the flexibility and versatility of decoder-only models seem to make Google's insistence on this direction less promising.

Task 3d

Look at your answers for task 3b. In each one the verb is modified by the passive voice and one at least one other type of tense information.

Verb Phrase	Tense
a) …have been gradually dominating…	Perfect tense (plus the adverb 'gradually')
b)	
c)	
d)	
e)	

Task 3e

Read the following short extract about the two main types of language models which is based on next week's lecture and answer the questions below.

Masked Language Model

Researchers train masked language models to predict to predict missing words in a sentence by leveraging context from both before and after the missing word. Companies commonly use masked language models for tasks such as sentiment analysis, classification, and code debugging, where understanding the overall context is important. Computer scientists base the design of masked language models on the Encoder submodule of a transformer architecture.

Autoregressive Language Model

Researchers train autoregressive language models to predict the next word in a sentence using only the previous words as context. Companies commonly use autoregressive language models for generating text one token at a time in tasks, such as creative writing. Computer scientists primarily base autoregressive language models on the Decoder submodule of a transformer architecture

1. How many times do the phrases masked language model and autoregressive language model appear in the text?
2. What words or shorter phrases could be used instead of these long phrases?
3. Do only researchers train models? Do only companies use them? Is everyone who works in AI design a computer scientist?

Task 3f

Work in pairs. Each student will rewrite one of paragraphs from **Task 3e** using the passive voice in every sentence. In each sentence a) remove the people or organization doing the action and b) use an alternative to 'masked language model'? When you have both finished compare your answers.

Pronunciation Practice – Consonant Clusters

A consonant cluster is when two or three consonant sounds join together in an English word. These are the possible two consonant clusters at the start of words in English, excluding the sound /s/. There are many more possibilities in the middle or end of words.

	/p/	**/t/**	**/k/**	**/b/**	**/d/**	**/g/**	**/m/**	**/n/**	**/f/**	**/v/**	**/θ/**	**/ʃ/**	**/h/**
+/l/	play	-	class	black	-	glass	-	-	fly	-	-	-	-
+/r/	pray	Trip	crime	brown	drop	grow	-	-	fry	-	three	shrink	-
+/w/	-	twins	queen	-	dwell	-	-	-	-	-	-	-	-
+/j/	pure	Tube	queue	beauty	due	-	music	news	few	view	-	-	huge

The sound /s/ is particularly common at the beginning of consonant cluster and joins with many other sounds that are not otherwise found in a cluster.

Sphere +/f/	slow +/l/	swim +/w/
snow +/n/	/S/	spot +/p/
smile +/m/	sky +/k/	star +/t/

The sound /s/ is also the only sound which can come at the start of a three-consonant cluster

	/sp/	**/st/**	**/sk/**
+/l/	**Splash**	**-**	**-**
+/r/	**Spray**	**Straw**	**Scream**
+/w/	**-**	**-**	**Squeak**
+/j/	**-**	**Stew**	**Skewer**

When speaking make sure you don't…

- Replace a consonant sound with a different one.
  For example: saying 'present' for 'pleasant' or saying 'queue' for 'crew'
- Leave out one of the consonant sounds.
  For example: saying 'poblem' for 'problem' or saying 'foo' for 'few'
- Add an extra vowel between consonants.
  For example: saying 'tewin' for 'twin' or saying 'faree' for 'free'

Task 4a – Listening and Discriminating.

*Listen to your teacher **(or record?)** and write down if they are saying the words from column A or column B.*

Column A	Column B
Blow	Below
Twin	To win
Slight	Sight
Playing	Praying
Glass	Grass
Three	Tree
Back	Black
Fry	Fly

Task 4b – Single Word Practice.

Now work in pairs. One student says a word from the pairs above and the other student try to identify if it is from column A or column B.

Task 4c – Sentence Practice.

Toss a coin. Read out the sentence as naturally as possible but making sure to pronounce the consonant cluster clearly.

Heads	Tails
Just blow your nose.	Just below your nose.
The girl is John's twin.	The girl is John's to win.
It was a terrible slight. (note: slight = insult)	It was a terrible sight.
I saw the child playing.	I saw the child praying.
The glass will be cut.	The grass will be cut.
That Chinese character means 'three'.	That Chinese character means 'tree'.
The dog is back.	The dog is black.
The bird will fry.	The bird will fly.

This section adapted from *Hewings, M. (2007) English Pronunciation in Use – Advanced. Cambridge: Cambridge University Press.*